

A microscopic image of cells, likely a histological section, showing various cell types and structures. The image is overlaid with a dark teal color and contains white text.

Cells II

BIOL2251

Chapter 3

Sections 3.3-3.4, 3.8-3.11



DNA Organization

DNA Organization

- **Instructions**

- I need 4 volunteers (2 pairs)
- Use your bodies to demonstrate the progression:
 - **Relaxed chromatin → condensed chromatin → chromosome → sister chromatids joined at centromere**
- **Bloom's taxonomy = Remember, Understand**

DNA Organization

- **Chromatin (relaxed):** DNA wrapped around histone proteins like beads on a string; found during interphase when DNA is being read/copied; looks like tangled thread under microscope
- **Chromatin (condensed):** Same DNA-histone complex but tightly coiled; happening as cell prepares for division
- **Chromosome:** Fully condensed chromatin visible during cell division; one continuous DNA molecule with associated proteins; X-shape seen in diagrams is actually TWO sister chromatids
- **Sister chromatids:** Two identical copies of a chromosome (made during S phase) joined at centromere; will separate during mitosis so each daughter cell gets one copy



Nucleus Command Center

Nucleus Command Center

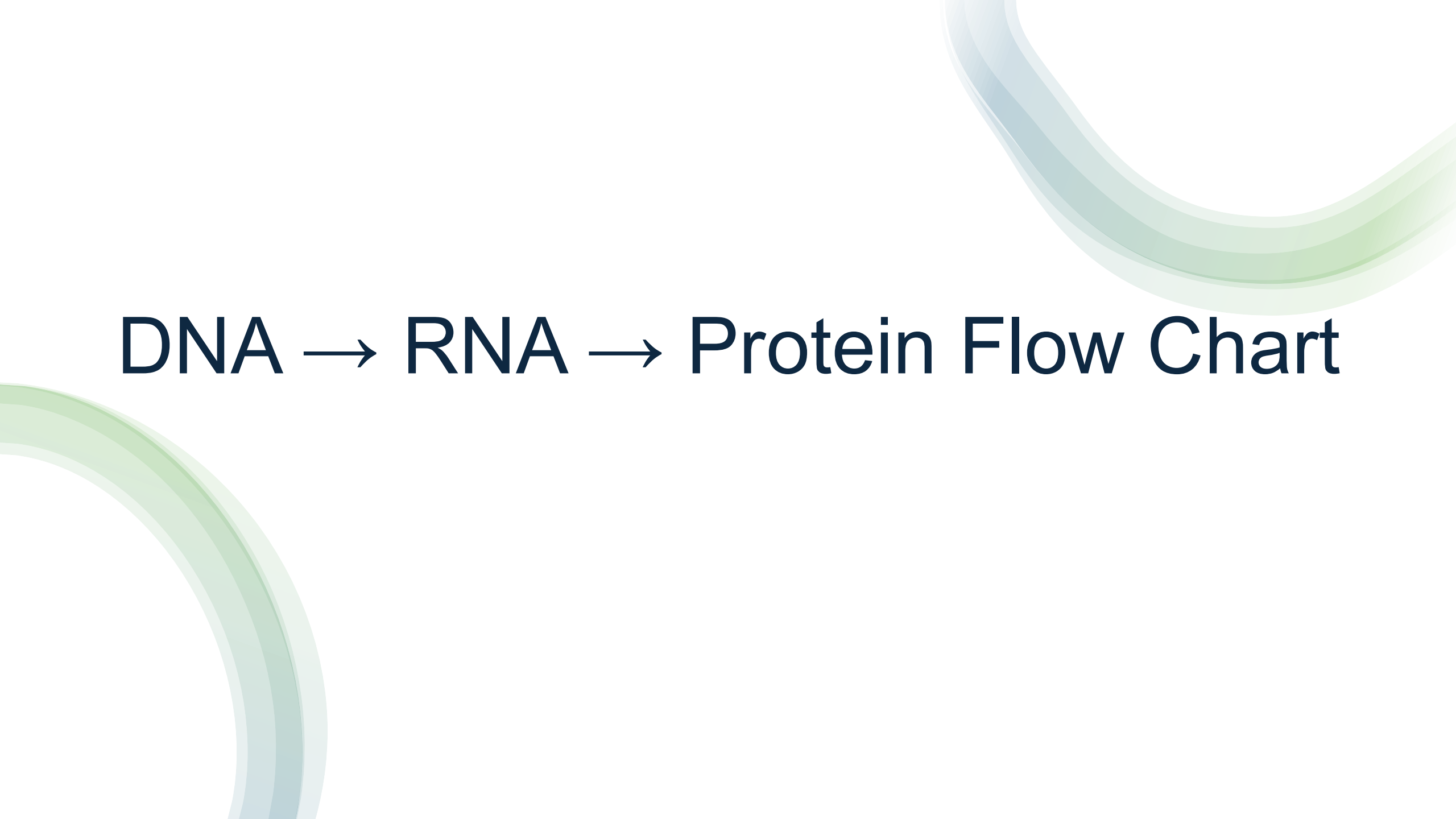
- **Instructions**

- Think about this analogy on your own: **“The nucleus is like a company headquarters.”**
- **Describe how DNA acts as the instruction manual, RNA as the messengers, and the nuclear envelope as security**
- Take 5 minutes, then discuss with a peer

- **Bloom’s taxonomy = Understand, Apply**

Nucleus Command Center

- **DNA = Master instruction manual/blueprint:** Contains all genetic information (genes) with instructions for making every protein the cell needs; stored in the nucleus
- **RNA = Messenger:** Carries copies of instructions (genes) from nucleus to cytoplasm where proteins are made; temporary working copies
- **Nuclear envelope = Security checkpoint:** Double membrane with pores that control what enters/exits; protects DNA from damage
- **Nuclear pores = Security gates:** Allow RNA and small molecules through but keep DNA inside and unwanted materials out



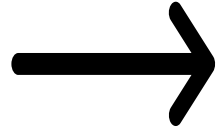
DNA → RNA → Protein Flow Chart

DNA → RNA → Protein Flow Chart

- **Instructions**

- Form smalls groups (1 group per table)
- **Create a flowchart showing the journey from DNA to protein**
- List the main processes and where the steps occur in the cell
- Think about the following: **transcription, translation, mRNA processing, mRNA, tRNA, ribosome, rough ER, amino acid, nucleus, and cytosol**
- Work for 10 minutes, then compare with a neighboring group
- **Bloom's taxonomy = Understand, Apply, Analyze**

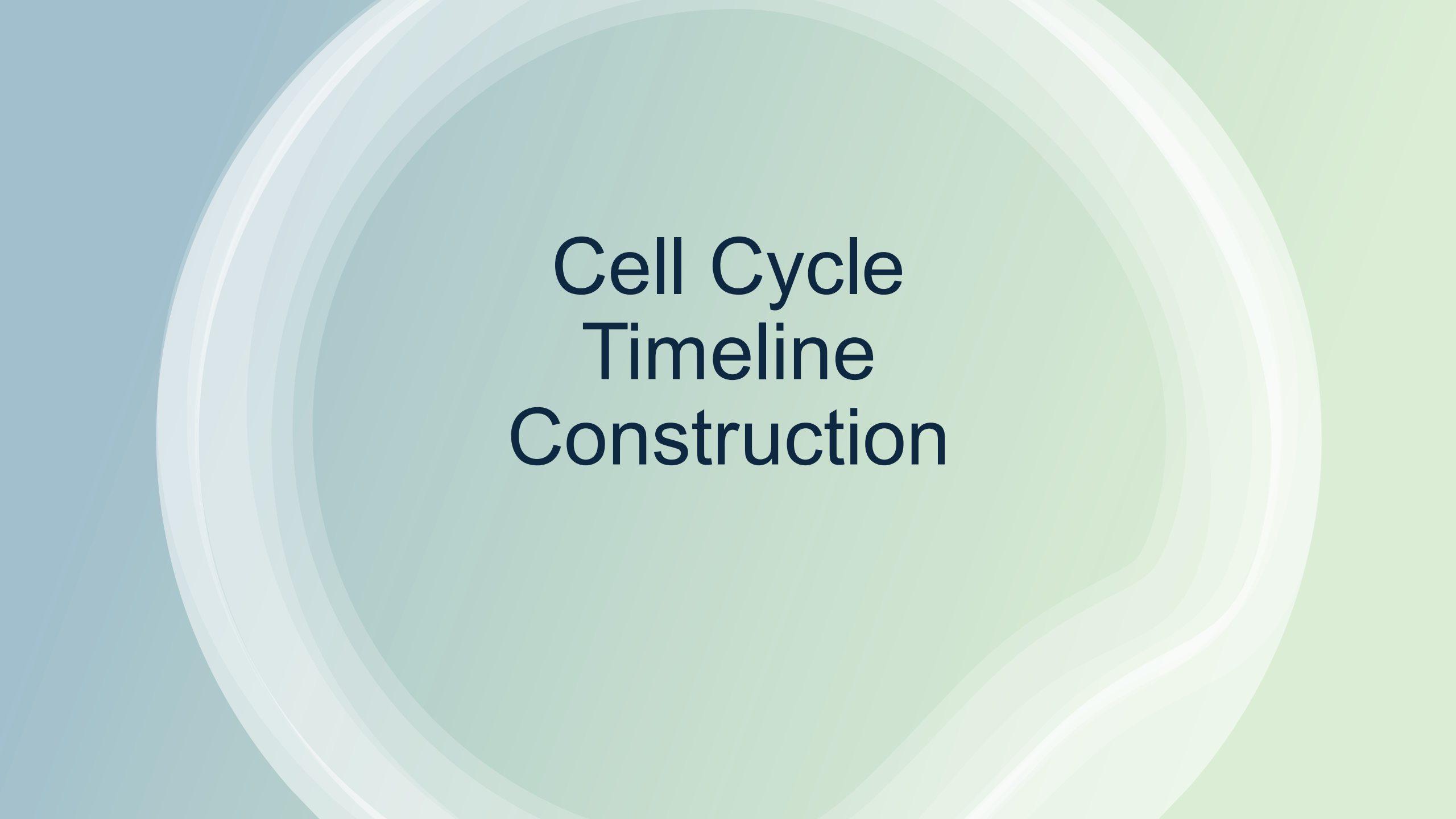
DNA



RNA



Protein



Cell Cycle Timeline Construction

Cell Cycle Timeline Construction

- **Instructions**

- Begin by working on your own
- **Draw a timeline of the cell cycle**
 - **Label G1, S, G2, and M phase (mitosis + cytokinesis)**
- For each phase, **write 1-2 key events**
- Take ~10 minutes to work and then compare with a partner
 - We will debrief as a class

- **Bloom's taxonomy = Remember, Understand**

Cell Cycle Timeline Construction

INTERPHASE (90% of cell life):

G1 (Gap 1) - First Growth Phase

- Cell grows in size
- Accumulates nutrients and energy (ATP)
- Produces more organelles (mitochondria, ribosomes, ER, etc.)
- Performs its normal, specialized functions
- Checkpoint: Cell checks if conditions are favorable to divide

Cell Cycle Timeline Construction

S (Synthesis)

- DNA replication occurs
- Each chromosome is duplicated → creates sister chromatids joined at centromere
- Histone proteins also synthesized
- Centrosomes duplicate
- Result: Cell now has 2× the DNA (46 chromosomes → 92 chromatids in humans)

Cell Cycle Timeline Construction

G2 (Gap 2) - Second Growth Phase

- Cell continues to grow
- Produces proteins needed for mitosis (tubulin for spindle fibers)
- Organelles duplicate
- Checkpoint: Cell verifies DNA was copied correctly and is ready for mitosis

Cell Cycle Timeline Construction

MITOSIS = Division of nucleus

1. Prophase

- Chromatin condenses into visible chromosomes (sister chromatids joined)
- Centrioles move to opposite poles
- Spindle fibers begin forming from centrosomes
- Nuclear envelope begins to break down
- Nucleolus disappears

Cell Cycle Timeline Construction

2. Pro-metaphase (sometimes combined with prophase)

- Nuclear envelope completely fragments
- Spindle fibers attach to kinetochores (protein structures on centromeres)
- Chromosomes begin moving toward cell center

Cell Cycle Timeline Construction

3. Metaphase

- Chromosomes align at cell equator (metaphase plate)
- Each chromosome attached to spindle fibers from both poles
- Checkpoint: Cell ensures all chromosomes properly attached before proceeding

Cell Cycle Timeline Construction

4. Anaphase

- Sister chromatids separate and move to opposite poles
- Pulled by spindle fiber shortening
- Cell elongates
- (Now each chromatid is considered an individual chromosome)

Cell Cycle Timeline Construction

5. Telophase

- Chromosomes arrive at poles and begin de-condensing back into chromatin
- Nuclear envelopes reform around each set of chromosomes
- Spindle fibers disappear
- Nucleoli reappear
- Results in 2 nuclei in 1 cell

Cell Cycle Timeline Construction

CYTOKINESIS = Division of cytoplasm

- Overlaps with late anaphase/telophase
- Cleavage furrow forms (contractile ring of microfilaments pinches cell in two)
- Organelles distributed to daughter cells
- Results in two separate daughter cells, each with one nucleus
- Each daughter cell is genetically identical to parent (same DNA)



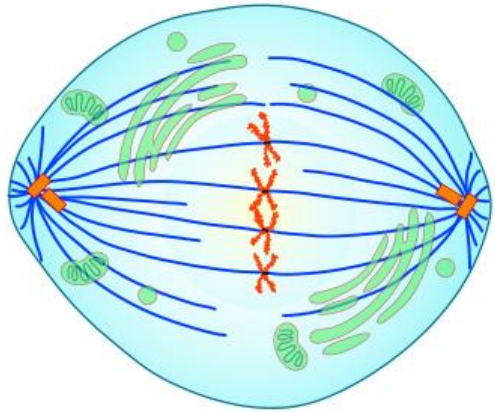
Mitosis Phase Identification

Mitosis Phase Identification

- **Instructions**

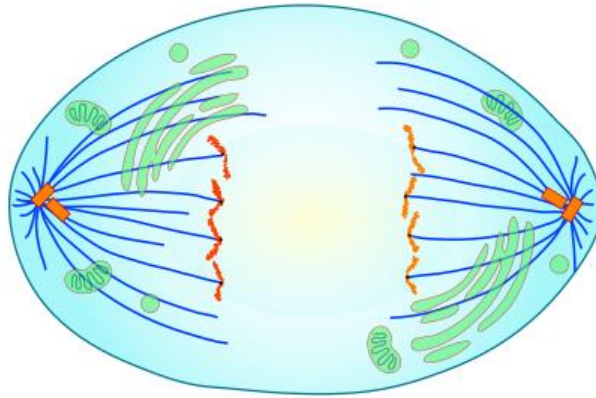
- Form small groups (4-6 people)
- I will present a set of unlabeled images showing the **stages of mitosis**
- **Identify each stage of mitosis and name one key event for each stage**
- We will work across groups until all stages are identified
- **Bloom's taxonomy = Remember, Apply**

A



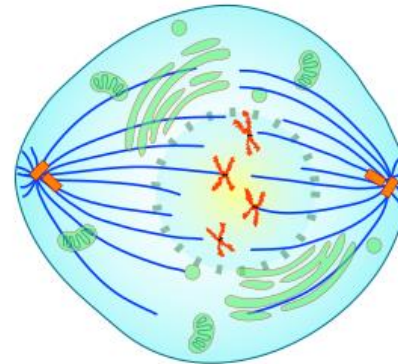
Metaphase

B



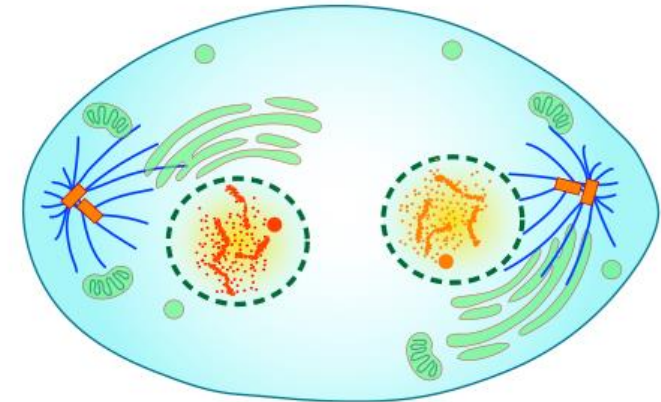
Anaphase

C



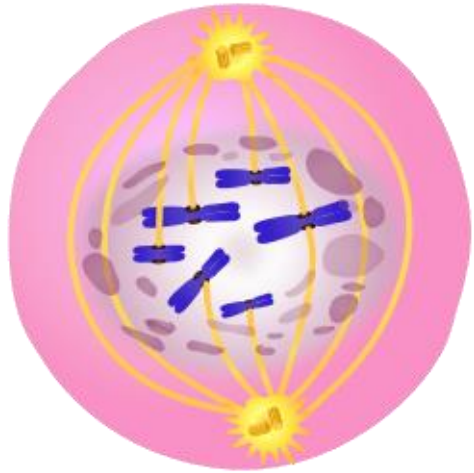
Prometaphase

D



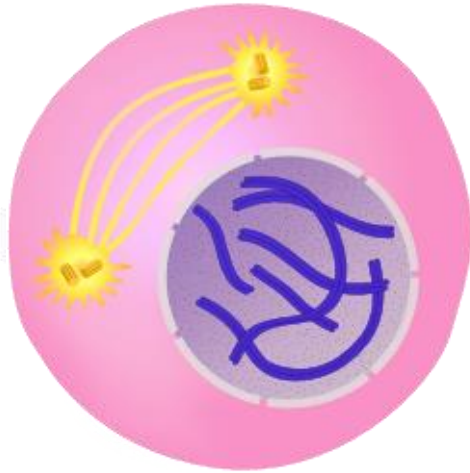
Telophase

A



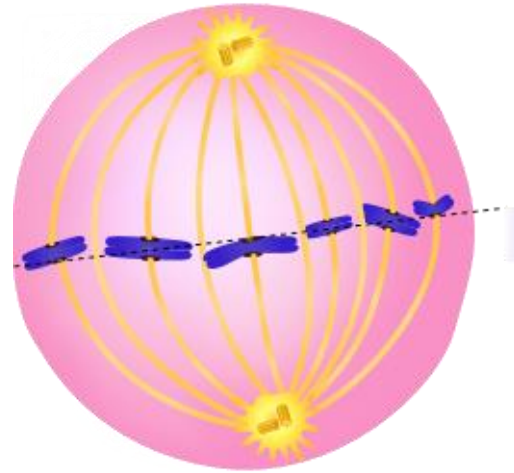
Prometaphase

B



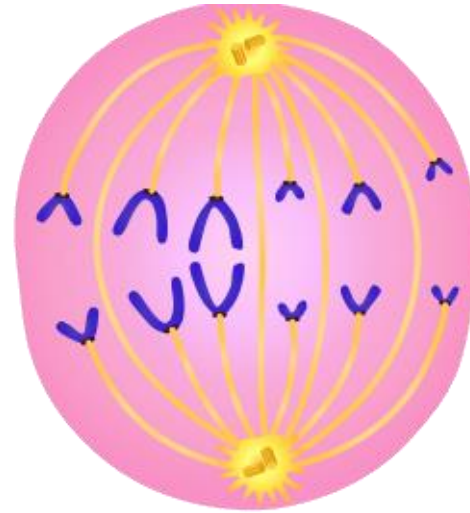
Prophase

C



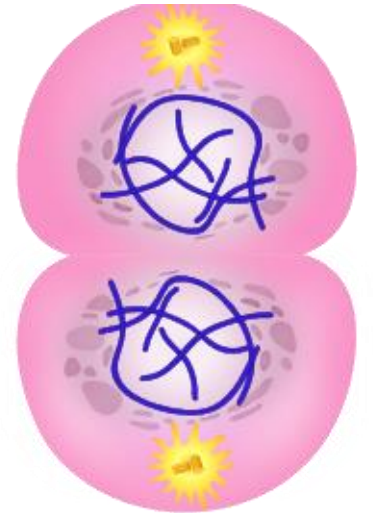
Metaphase

D



Anaphase

E



Telophase +
Cytokinesis

Mitosis Phase Identification

Interphase (for contrast):

- Nucleus intact with nuclear envelope visible
- Chromatin appears as diffuse, grainy material (not distinct chromosomes)
- Nucleolus visible

Prophase:

- Chromosomes visible as distinct X-shapes or thread-like structures (condensing)
- Centrioles visible moving toward poles

Pro-metaphase:

- Nuclear envelope breaking down or fragmented
- Chromosomes fully condensed
- Chromosomes scattered, not yet aligned
- Spindle fibers visible

Mitosis Phase Identification

Metaphase:

- Chromosomes lined up in middle of cell (single line across center)
- No nuclear envelope
- Cell typically looks circular/symmetric

Anaphase:

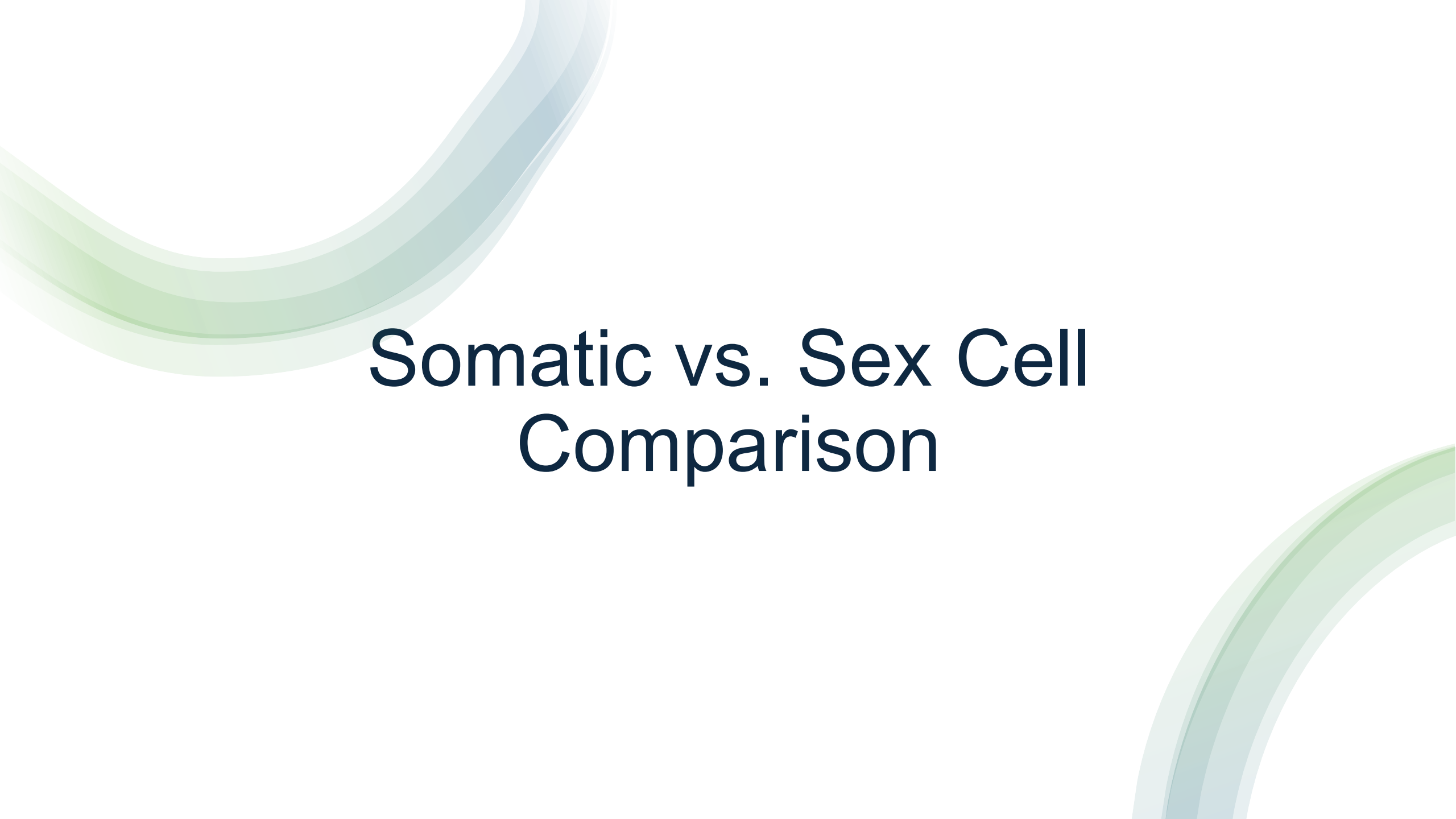
- Sister chromatids separated and moving to opposite ends
- V-shape or pulled appearance
- Cell may be elongating

Telophase:

- Two distinct groups of chromosomes at opposite poles
- Chromosomes begin to de-condense (look less distinct)
- Cleavage furrow visible (cell pinching)
- Nuclear envelopes reforming

Cytokinesis:

- Two nearly separate cells
- Each with its own nucleus
- Still connected by thin bridge of cytoplasm OR fully separated

The background features decorative curved lines in shades of green and blue, sweeping across the top and bottom of the slide.

Somatic vs. Sex Cell Comparison

Somatic vs. Sex Cell Comparison

- **Instructions**

- Work in pairs or small groups
- **Create a Venn diagram to compare somatic and sex cells (gametes)**
- Make sure to include **chromosome number, type of division, where found, function, and one example**
- Compare with a neighboring pairs or small group

- **Bloom's taxonomy = Understand, Analyze**

Somatic vs. Sex Cell Comparison

SOMATIC CELLS (body cells):

- **Chromosome number:** Diploid ($2n = 46$ in humans) - two sets of chromosomes (one from each parent)
- **Type of division:** Mitosis
- **Where found:** Everywhere in body except reproductive organs (testes/ovaries)
- **Function:** Growth, repair, maintenance of body
- **Examples:** Skin cells, muscle cells, liver cells, nerve cells, blood cells

Somatic vs. Sex Cell Comparison

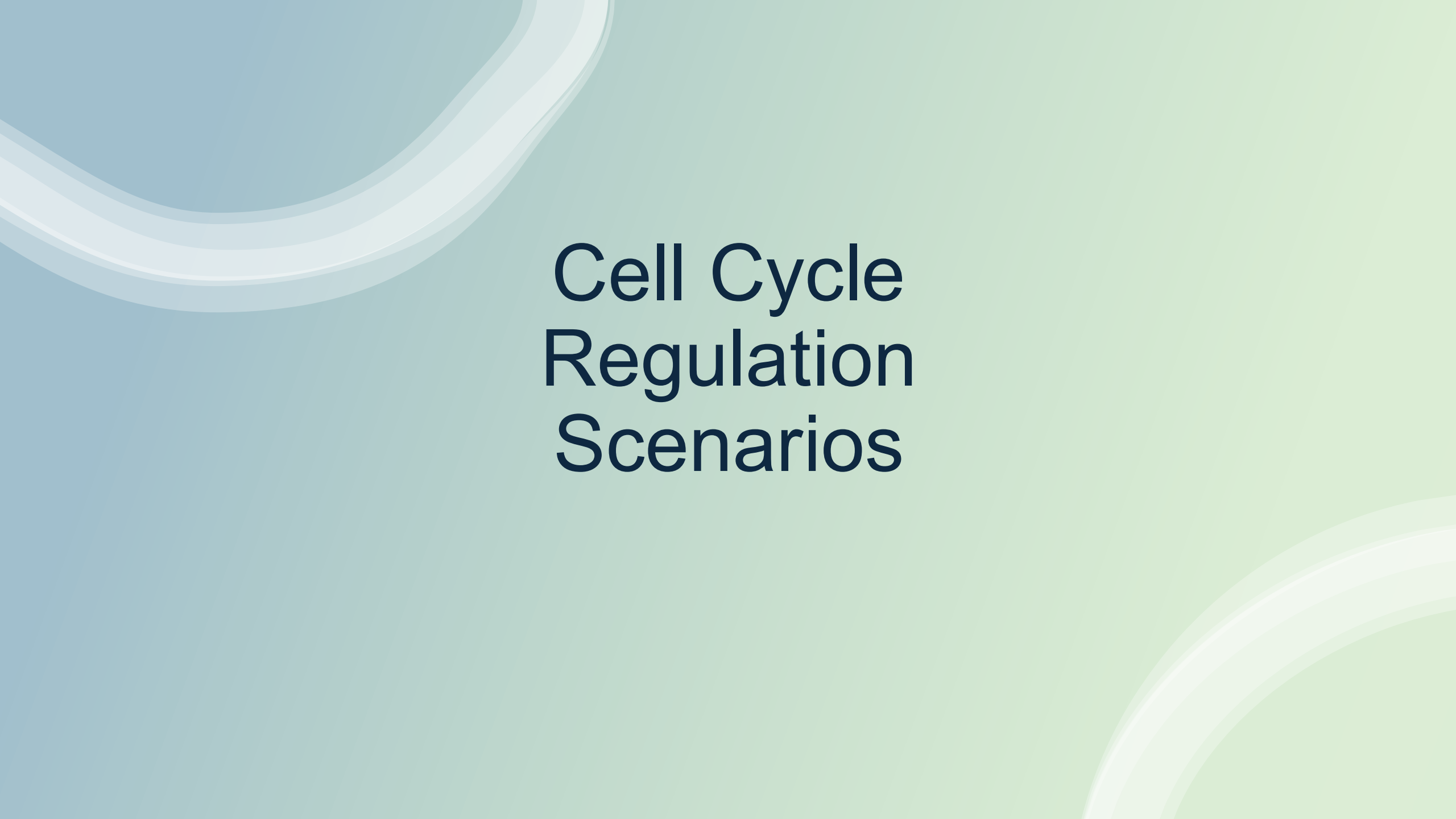
SEX CELLS (gametes):

- **Chromosome number:** Haploid ($n = 23$ in humans) - one set of chromosomes
- **Type of division:** Meiosis (not mitosis)
- **Where found:** Only produced in testes (sperm) or ovaries (eggs)
- **Function:** Sexual reproduction - fuse during fertilization to create new organism
- **Examples:** Sperm (male), eggs/ova (female)

Somatic vs. Sex Cell Comparison

BOTH:

- Made of cells with DNA
- Have chromosomes
- Can divide
- Perform cellular processes (respiration, protein synthesis, etc.)



Cell Cycle Regulation Scenarios

Cell Cycle Regulation Scenarios

- **Instructions**

- Work in small groups (4-6 people)
- You will work through different types of cells
 - **Skin wound healing, mature cardiac muscle fiber, liver tissue after partial removal, cancer cell, normal adult skin cell**
- **Predict if the growth rate would be low, medium, or high**
- Try to think of factors that affect the mitotic rate
- Groups will share during whole class debrief
- **Bloom's taxonomy = Understand, Apply, Analyze**

Cell Cycle Regulation Scenarios

Scenario 1: Skin wound healing

- **Mitosis rate:** High
- **Regulation factors:**
 - Growth factors released from damaged tissue stimulate division
 - Contact inhibition removed (cells no longer surrounded by neighbors)
 - Chemical signals (cytokines) promote rapid cell division
 - Stops when wound is covered (contact inhibition restored)

Scenario 2: Mature cardiac muscle fiber (heart)

- **Mitosis rate:** Stopped (in G0)
- **Regulation factors:**
 - Highly specialized cells that permanently exited cell cycle
 - Cell cycle inhibitors prevent division
 - This is why heart attacks cause permanent damage

Cell Cycle Regulation Scenarios

Scenario 3: Liver tissue after partial removal

- **Mitosis rate:** High
- **Regulation factors:**
 - Liver has remarkable regenerative capacity
 - Growth factors and hormones stimulate division
 - Continues until liver reaches normal size, then stops
 - Contact-dependent inhibition restored at normal size

Scenario 4: Cancer cell

- **Mitosis rate:** High (uncontrolled)
- **Regulation factors (what's broken):**
 - Mutations in genes that control cell cycle
 - Ignores growth factor signals
 - No contact inhibition
 - Ignores apoptosis (cell death) signals
 - Checkpoints not functioning

Cell Cycle Regulation Scenarios

Scenario 5: Normal adult skin cell

- **Mitosis rate:** MODERATE/CONTROLLED
- **Regulation factors:**
 - Divides to replace old/dead cells at surface
 - Contact inhibition working (slows when surrounded)
 - Growth factors present in normal amounts
 - Cell cycle checkpoints functioning
 - Telomeres limiting lifespan

General factors affecting mitosis rate:

- **Growth factors:** Chemical signals that promote division
- **Nutrient availability:** Cells need resources to divide
- **Cell size:** Must reach certain size before division
- **DNA integrity:** Checkpoints halt division if DNA damaged
- **Contact inhibition:** Cells stop dividing when surrounded by other cells
- **Telomere length:** Limits number of divisions
- **Cyclins and CDKs:** Internal proteins that control cycle progression



Cancer Mini Cases

Cancer Mini Cases

- **Instructions**

- Work in small groups (4-6 people)
- For your assigned scenario, discuss the following:
 - **What effect would this have on the cell cycle?**
 - **How might this change lead to cancer?**
- Groups for each scenario will share their thinking and reasoning
- We will regroup and debrief as class

- **Bloom's taxonomy = Understand, Analyze, Evaluate**

Cell Cycle Checkpoints

- **G1 checkpoint (before DNA replication):** Is the cell large enough? Is DNA intact? Are growth signals present?
- **G2 checkpoint (before mitosis):** Was DNA copied correctly? Any damage?
- **M checkpoint (during mitosis):** Are all chromosomes attached properly to spindle fibers before separation?

Regulatory Proteins

- **Cyclins and cyclin-dependent kinases (Cdks):** Work together like a lock and key to push the cell through checkpoints
- **Tumor suppressors (like p53):** Act as brakes – stop the cell cycle if DNA is damaged, and trigger repair or cell death if needed
- **Proto-oncogenes:** Normal “gas pedals” that encourage division when appropriate (e.g., growth factors)
 - If mutated, they can become oncogenes, stuck in the “on” position

Cancer Mini Cases

Case A

- Mutation in p53
- The “brakes” fail at the G1/G2 checkpoint.
- **Question: What happens if DNA is damaged but the cell keeps dividing?**

Case B

- Overactive Growth Factor Receptor
- The “accelerator” is stuck on. The cell keeps getting signals to divide even without external cues.
- **Question: How does this affect cell division and risk for tumor formation?**

Case A: Mutation in p53

- If p53 is not working, cells with damaged DNA are not stopped at the checkpoint
- Cells replicate with errors, which means mutations accumulate
- Over time, this makes it more likely the cells will grow uncontrollably and form a tumor
- **Key point:** Damaged DNA + no checkpoint control = mutation build-up → cancer risk increases.

Case B: Overactive Growth Factor

- Cells will keep dividing even when the body doesn't need new cells
- This can lead to too many cells being produced, creating a mass (tumor)
- If other mutations occur, this mass can become malignant and spread
- **Key point:** Uncontrolled “go” signals = excessive division → tumor growth.